

Functional Annotation of PP_4452 (Q88EL0) in *Pseudomonas putida* KT2440

Target: Gene PP_4452 / UniProt Q88EL0 / *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / **KT2440**), organism code PSEPK. **Protein:** 359 aa, soluble cytoplasmic NAD(P)-dependent oxidoreductase. **Report date:** 2026-07-18

1. Summary (Answer to the Research Question)

PP_4452 encodes a **cytoplasmic NAD(P)-dependent oxidoreductase of the opine/lysopine dehydrogenase family** (an "opine dehydrogenase," EC 1.5.1.28-type; KEGG ortholog K04940). Its most likely primary reaction is the reversible NAD(P)(H)-dependent interconversion between an "**opine**" (an **N-(carboxyalkyl)-amino acid / secondary-amine conjugate**) and its **constituent amino acid + 2-oxo (keto) acid** — i.e., reductive condensation in the anabolic direction, or oxidative cleavage in the catabolic direction. The gene lies at the head of a **complete opine-utilization operon** (opine ABC importer + a two-subunit octopine oxidase OoxAB + a LysR regulator), which strongly indicates a **catabolic role in the uptake and breakdown of opine growth substrates** in this soil/rhizosphere bacterium. The enzyme works in the **cytoplasm**.

Important annotation caveat. The UniProt name "2-dehydropantoate 2-reductase / ketopantoate reductase (EC 1.1.1.169)" attached to Q88EL0 is an **automated (ARBA) electronic prediction, not an experimental or manually curated assignment, and is almost certainly a mis-annotation**. It arises because ketopantoate reductase (KPR/PanE) and opine dehydrogenases share the same two-domain protein fold. Family-level classifiers, KEGG orthology, genomic context, and the presence of three *bona fide* ketopantoate reductases elsewhere in the genome all indicate PP_4452 is **not** the pantothenate-pathway KPR. This report treats PP_4452 as an opine dehydrogenase and explains the evidence in full.

No experimental study of PP_4452 itself exists; the functional assignment is based on rigorous bioinformatic and evolutionary inference plus the biochemistry of characterized homologs.

2. Gene/Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88EL0	UniProtKB
Locus tag	PP_4452 (GenBank AAN70027)	EMBL AAN70027.1
Organism	<i>P. putida</i> KT2440 (PSEPK)	UniProt
Length / location	359 aa; genome complement 5,051,434–5,052,513	KEGG ppu:PP_4452
UniProt RecName (automated)	2-dehydropantoate 2-reductase, EC 1.1.1.169	ARBA (ECO:0000256)
KEGG ortholog	K04940 opine dehydrogenase (EC 1.5.1.28)	KEGG
GenBank product	"NAD/NADP octopine/nopaline dehydrogenase family protein"	KEGG/GenBank
eggNOG / OrthoDB	COG0240 / 6135265at2	UniProt xrefs

The locus tag, organism, and accession all match the target. **The disagreement is purely at the level of assigned function** (KPR vs. opine dehydrogenase), which Section 3 resolves. The gene is genuine and correctly identified; the literature below concerns the protein *family*, since no paper addresses PP_4452 directly.

3. Resolving the Function: Opine Dehydrogenase, Not Ketopantoate Reductase

Ketopantoate reductase (KPR) and opine dehydrogenases are **structural homologs**: both are built from an N-terminal Rossmann NAD(P)(H)-binding domain plus a C-terminal α -helical domain, and KPR is a member of the 6-phosphogluconate-dehydrogenase C-terminal-domain-like superfamily [PMID 11724562]. This shared fold is exactly what causes automated pipelines to cross-label members of one family as the other. Five independent lines of evidence show PP_4452 belongs to the **opine dehydrogenase** side of this fold:

- 1. Family-level classification (strongest signal).** InterPro assigns Q88EL0 to **family IPR051729 "Opine/Lysopine Dehydrogenase,"** and PANTHER independently assigns **family PTHR38015 "Opine/Lysopine Dehydrogenase."** In InterPro, family membership is the functional classification; the ketopantoate-reductase match (IPR013332 "KPR N-terminal domain," Pfam PF02558 PanE/ApbA) is only a **domain-level** hit reflecting the shared Rossmann fold.
- 2. Diagnostic domain and structural signatures.** Pfam also matches **PF02317 "NAD/NADP octopine/nopaline dehydrogenase, α -helical domain,"** and the CATH homologous superfamily for the C-terminal domain is **G3DSA:1.10.1040.10 "N-(1-D-carboxylethyl)-L-norvaline dehydrogenase"** (CENDH, a characterized bacterial opine dehydrogenase).
- 3. KEGG orthology.** PP_4452 is assigned **K04940 opine dehydrogenase (EC 1.5.1.28)** — not the KPR ortholog K00077.
- 4. The real KPRs are elsewhere.** *P. putida* KT2440's genuine pantothenate-pathway ketopantoate reductases are **PP_1351, PP_2325, and PP_2998**, all annotated K00077 / EC 1.1.1.169. PP_4452 is therefore redundant to and distinct from the housekeeping KPR — it is not needed for vitamin B5/CoA synthesis.
- 5. Protein length.** PP_4452 is **359 aa**, matching the opine dehydrogenase family (e.g., *Arthrobacter* sp. 1C opine DH, UniProt Q44297 = 359 aa; *Pecten maximus* octopine DH $\approx$ 399 aa) and clearly longer than single-domain KPR (*E. coli* PanE = 303 aa).
- 6. Quantitative pairwise identity** (global BLOSUM62 alignment, this work). PP_4452 is **most similar to a characterized bacterial opine dehydrogenase** — *Arthrobacter* sp. 1C opine DH (Q44297): **29.6% identity / 49.3% similarity over a near-full-length 351-residue alignment** — and to *P. aeruginosa* pseudopaline synthase CntM (Q9HUX5): 24.6% / 42.4% over 354 aa. By contrast it aligns to *E. coli* ketopantoate reductase (P0A9J4) at only **22.7% identity over 295 aa**, meaning **~64 residues of PP_4452 have no KPR counterpart** — only the shared N-terminal Rossmann region aligns. Thus PP_4452 matches opine

dehydrogenases along essentially its entire length but matches KPR only in the common cofactor-binding subdomain.

Conclusion: The evidence overwhelmingly favors an **opine/lysopine dehydrogenase**. The KPR RecName should be regarded as a low-confidence electronic artifact.

4. Primary Function and Reaction Chemistry

4.1 Reaction class

Opine dehydrogenases (OpDHs) catalyze the **NAD(P)(H)-dependent reductive condensation of an amino acid and a 2-oxo acid into a secondary-amine "opine,"** and the reverse oxidative cleavage. The archetype is octopine dehydrogenase (OcDH) from the scallop *Pecten maximus*, which "catalyzes the reductive condensation of L-arginine and pyruvate to octopine ... [and] oxidizes glycolytically born NADH to NAD⁺" [PMID 18599075]. Generic scheme:



The hydride is delivered from/accepted by the nicotinamide cofactor bound in the N-terminal Rossmann domain; the reaction creates (or breaks) a C–N bond and generates a new chiral center.

4.2 Substrate specificity

The **exact opine substrate of PP_4452 has not been experimentally determined**. In characterized OpDHs, substrate identity is set by **size selection in the active site** — a "molecular ruler" that combines electrostatic recognition of the amino-acid α -carboxylate/side chain with steric gating of the 2-oxo-acid moiety [PMID 18599075]. Family members differ widely in their amino-acid partner (arginine → octopine/nopaline; ornithine/lysine → various; norvaline → CENDH product; glutamate → staphylopine/pseudopaline-type). Based on family and genomic context, PP_4452 is predicted to act on an **opine-type N-(carboxyalkyl)amino acid**; defining the precise pair requires enzymology.

4.3 Cofactor

NAD(P)(H)-dependent (the family uses NADH or NADPH depending on the member). The intact N-terminal glycine-rich dinucleotide-binding motif is present near the N-terminus of PP_4452 (...**GIIGAGGIG**..., residues ~10–18), confirming a functional Rossmann cofactor site.

5. Structural Basis

- **Fold:** two-domain architecture — N-terminal **Rossmann NAD(P)(H)-binding domain** (CATH 3.40.50.720 / SSF51735) + C-terminal **α -helical substrate/dimerization domain** (6-phosphogluconate-dehydrogenase-like; SSF48179, CATH 1.10.1040.10). This is the canonical opine-dehydrogenase/KPR superfamily fold [PMID 11724562].
- **Catalytic logic (from homologs):** catalysis occurs in the interdomain cleft; cofactor binds in the Rossmann lobe and the amino-acid/2-oxo-acid substrates bind in the cleft, with domain closure bringing catalytic residues into register. In the homologous KPR this closure repositions a catalytic Lys/Glu pair over the substrate [PMID 17229734,

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opine dehydrogenases use an analogous interdomain active site with substrate-recognition residues that impose the "molecular ruler" specificity [PMID 18599075].

- An **AlphaFold model (AlphaFoldDB Q88EL0)** is available and is expected to reproduce this bilobed fold, offering a route to computational substrate docking.

6. Subcellular Localization

PP_4452 is a **soluble cytoplasmic protein**. A Kyte–Doolittle hydropathy scan (19-residue window, this work) gives a maximum local hydropathy of only **1.36**, below the ~1.6–1.9 threshold for a membrane-spanning helix, so there is **no predicted transmembrane segment**; the N-terminus begins directly with the glycine-rich Rossmann motif (MEMQSYKVG**GIIGAGGIG**...) rather than a cleavable signal peptide. NAD(P)-dependent dehydrogenases of this family act in the cytosol. This is consistent with its pathway role: the

substrate (an opine) is first **imported across the inner membrane by the co-encoded ABC transporter** (membrane-bound permeases + periplasmic/lipoprotein substrate-binding protein), after which PP_4452 processes it **intracellularly**.

7. Pathway Context: A Complete Opine-Utilization Operon

PP_4452 is the first gene of a contiguous, co-oriented (minus-strand) gene cluster spanning ~5.051–5.061 Mb that encodes a full opine uptake-and-catabolism system:

Locus	KEGG KO	Annotated product	Role
PP_4452	K04940	opine dehydrogenase (EC 1.5.1.28)	NAD(P)-dependent opine dehydrogenase (this gene)
PP_4453	K02031	ABC transporter ATP-binding	Importer — ATPase
PP_4454	K02034	"Opine ABC transporter" permease	Importer — permease
PP_4455	K02033	ABC permease	Importer — permease
PP_4458	K02035	ABC substrate-binding protein	Importer — solute-binding subunit
PP_4456	K24954	octopine oxidase subunit A (OoxA)	Flavin/Fe-S opine-catabolic oxidase
PP_4457	K24955	octopine oxidase subunit B (OoxB)	Flavin/Fe-S opine-catabolic oxidase
PP_4460	—	LysR-family transcriptional regulator	Substrate-responsive induction
PP_4461	—	Major facilitator superfamily transporter	Auxiliary transport
PP_4462	—	putative keto-acid (HMG/oxaloacetate) aldolase	Downstream product catabolism

Bacterial opine dehydrogenases function in **opine catabolism** — for example, "Opines, such as nopaline and octopine ... are specifically catabolized by opine dehydrogenase (OpnDH)" [PMID 40581604]. The classical multi-subunit OpnDH/opine oxidase "contain[s] FAD, FMN, and 2 iron-sulfur clusters as cofactors" [PMID 40581604], matching the two-subunit **OoxAB (PP_4456/PP_4457)** in this operon. Thus the cluster contains **two complementary opine-degrading activities** — the NAD(P)-linked opine dehydrogenase (PP_4452) and the flavin/Fe-S opine oxidase (OoxAB) — alongside a dedicated importer and a LysR regulator that likely induces the operon when an opine is available. The most parsimonious interpretation is that this locus lets *P. putida* **import and consume opine (N-carboxyalkyl-amino-acid) compounds as carbon and nitrogen sources**, with PP_4452 cleaving the opine into an amino acid + 2-oxo acid that feed central metabolism.

Distinction from the metallophore branch

The same superfamily includes **biosynthetic** "opine" dehydrogenases that build nicotianamine-like metallophores (staphylopin, pseudopaline, yersinopine) used by pathogens to counter host nutritional immunity [PMID 31300464]. Those enzymes (CntM-type) sit in *cnt* operons together with a **nicotianamine synthase (CntL)** and a dedicated metallophore exporter — none of which are present at the PP_4452 locus. This further supports a **catabolic**, nutrient-utilization role rather than metallophore biosynthesis.

8. Evidence Summary

Claim	Evidence type	Strength
Belongs to opine/lysopine dehydrogenase family (not KPR)	Bioinformatic: InterPro & PANTHER family assignment, CATH/Pfam signatures, KEGG orthology	Strong
Housekeeping KPRs are separate genes (PP_1351/2325/2998)	Comparative genomics (KEGG K00077)	Strong
Catalyzes NAD(P)-dependent opine $\Rightarrow$ amino acid + 2-oxo acid	Homology to characterized OpDHs [PMID 18599075]	Moderate–strong (by inference)
Functions in opine catabolism/uptake	Genomic context: complete importer + OoxAB oxidase + LysR [PMID 40581604]	Strong (contextual)
Cytoplasmic localization	Sequence features (no signal/TM) + family biochemistry	Strong
Exact opine substrate	—	Unknown (experimental gap)
Direct enzymology of PP_4452	—	None available

9. Supported and Refuted Hypotheses

- **REFUTED (as the primary function): Ketopantoate reductase / 2-dehydropantoate 2-reductase (EC 1.1.1.169).** The UniProt ARBA name is a fold-based electronic artifact; family classifiers, KEGG orthology, and the existence of three dedicated KPRs elsewhere argue against it. (A residual, unproven possibility that the enzyme retains weak promiscuous 2-oxo-acid reductase activity cannot be formally excluded without assays, but it is not the physiological role.)
- **SUPPORTED: NAD(P)-dependent opine/lysopine dehydrogenase (EC 1.5.1.28-type, K04940)** acting in the cytoplasm.

- **SUPPORTED: Catabolic role in an opine-utilization operon** (uptake + degradation of an opine growth substrate), based on the co-encoded ABC importer, OoxAB opine oxidase, and LysR regulator.
- **NOT SUPPORTED: Metallophore (staphylopine/pseudopaline-type) biosynthesis** — the diagnostic nicotianamine synthase (CntL) and metallophore exporter are absent.

10. Limitations and Future Directions

- **No direct biochemistry** exists for PP_4452; conclusions rest on bioinformatics, homology, and genomic context. High-throughput/automated database annotations conflict, and the UniProt RecName is misleading.
 - **The precise opine substrate and cofactor preference (NADH vs NADPH) are undetermined.** Priority experiments: (i) recombinant expression and an activity screen against candidate opines and amino-acid/2-oxo-acid pairs (arginine+pyruvate, ornithine/lysine/glutamate + pyruvate/ α -ketoglutarate); (ii) steady-state kinetics and cofactor specificity; (iii) AlphaFold-guided substrate docking and active-site "molecular-ruler" analysis; (iv) growth phenotyping of a PP_4452 (and operon) deletion on candidate opines as sole C/N source; (v) transcriptional induction tests to confirm LysR (PP_4460) control.
 - Confirming which of PP_1351/PP_2325/PP_2998 is the functional pantothenate-pathway KPR would complete the vitamin B5 picture but is outside this gene's scope.
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Key References

- Matak-Vinković et al. (2001) *Biochemistry* — E. coli ketopantoate reductase structure; shared 6PGD-like superfamily fold.

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- Zheng & Blanchard (2000) — KPR catalytic residues (Lys/Glu).

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- Ciulli et al. (2007) — KPR ternary complex, domain closure.

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- Smits et al. (2008) *J. Biol. Chem.* — octopine dehydrogenase structure; reaction and "molecular ruler" specificity.

[P 18599075](#)

- Laffont et al. (2019) — opine-type metallophores (staphylopin/pseudopaline/yersinopine).

[P 31300464](#)

- Watanabe et al. (2025) — bacterial opine catabolism; monomeric flavin opine dehydrogenase and multi-subunit OpnDH cofactors.

[P 40581604](#)

- Database sources: UniProt Q88EL0; KEGG ppu:PP_4452 and neighbors; InterPro/PANTHER/CATH/Pfam classifications.

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